

CUSTOMER RETENTION & CHURN ANALYSIS DASHBOARD

A Power BI Business Analytics & Data Visualization Project

Future Interns – Data Science & Analytics Internship
Task 2

Prepared by: Saniya Kalawant

Power BI | Business Analytics | Data Visualization

Project Overview

This project delivers an interactive Power BI Customer Churn Dashboard designed to analyze customer retention, identify the major drivers of churn, and provide actionable recommendations for reducing customer loss. The analysis focuses on understanding who is churning, why they may be churning, and where retention efforts should be prioritized.

Objectives

- Monitor overall customer churn
- Identify high-risk customer segments
- Understand key churn drivers
- Compare churn across contract types and payment methods
- Analyze churn across internet service types
- Understand the relationship between tenure and churn
- Develop targeted customer-retention strategies

Tools & Technologies

- Microsoft Power BI
- Power Query
- DAX
- Data Visualization
- Business Intelligence
- GitHub

Dataset, Data Preparation & KPIs

Dataset

The project uses a telecom customer churn dataset containing customer-level information such as contract type, payment method, internet service type, tenure, monthly charges, and churn status, enabling analysis of retention and churn behavior across customer segments.

Data Preparation

- Reviewed column data types and corrected inconsistencies
- Verified and cleaned churn status and tenure fields
- Grouped tenure into meaningful bands for trend analysis
- Standardized categorical fields such as contract type, payment method and internet service
- Used Power Query for data transformation
- Created DAX measures for churn, retention and segment-level KPIs

Key Performance Indicators (Overview Page)

KPI	Value
Total Customers	7,043
Churned Customers	1,869
Retained Customers	5,174
Churn Rate	26.5%
Monthly Revenue	\$456K
Avg. Monthly Charge	\$65

Additional KPIs tracked on the Churn Analysis page include:

- Avg. Monthly Charge – Churned Customers (\$74)
- Avg. Tenure – Churned Customers (18 Months)

Key Metrics & Formulas

Churn Rate = Churned Customers / Total Customers
Retention Rate = Retained Customers / Total Customers
Churned Customers = Number of customers where Churn = “Yes”

Dashboard Overview

Overview Page

The Overview page provides a high-level summary of customer retention and churn performance, including total, churned and retained customers, overall churn rate, and monthly revenue. Supporting visuals break down churn by contract type, gender and payment method, alongside overall customer status.



Figure 1: Overview

Churn Analysis

The Churn Analysis page focuses on identifying the major factors associated with customer churn, including payment method, partner status, internet service type, contract type and tenure. This page highlights where churn is concentrated and helps isolate the strongest churn drivers.



Figure 2: Churn Analysis

Insights & Action Plan

The dashboard's Insights & Action Plan page converts the analysis into business-focused conclusions and practical recommendations for improving customer retention.



Figure 3: Insights & Action Plan

Key Business Insights

■ New Customers Are at the Highest Risk

The 0–12 month customer group has the highest observed churn rate at 47.4%, declining consistently as tenure increases. The first year represents the most important retention window — early engagement, proactive support and personalized communication can help reduce customer loss.

Tenure Group	Churn Rate
0–12 Months	47.4%
13–24 Months	28.7%
25–36 Months	21.6%
37–48 Months	19.0%
49–60 Months	14.4%
60+ Months	6.6%

■ Month-to-Month Customers Have High Churn

Month-to-month customers show approximately 42.7% churn, considerably higher than customers on longer-term contracts. Contract commitment appears strongly associated with retention, and encouraging suitable customers toward longer-term plans could improve outcomes.

■ Fiber Optic Customers Show High Churn

Fiber optic customers churn at 41.9%, compared with 19.0% for DSL and 7.4% for customers without internet service. This should be investigated further, particularly around service quality, pricing, technical issues, customer expectations and competitive pressure.

Internet Service	Churn Rate
Fiber Optic	41.9%
DSL	19.0%
No Internet Service	7.4%

■ Payment Method Can Help Identify Risk

Electronic check customers show the highest churn rate among the payment methods analyzed. Payment behavior can be used as an additional customer-risk signal when developing churn-risk segments.

Risks & Recommendations

Business Risks

- New customers churn at nearly 7x the rate of long-tenured customers, exposing revenue during the first year.
- Month-to-month customers make up a disproportionate share of churn compared with contracted customers.
- Fiber optic customers, despite being a premium service segment, show the highest churn among internet service types.
- Electronic check customers are harder to retain than customers on automatic payment methods.

Actionable Recommendations

1. Strengthen new-customer onboarding through 30/60/90-day programs, welcome offers, proactive support, satisfaction surveys and personalized communication during the first 12 months.
2. Encourage longer-term contracts by targeting suitable month-to-month customers with annual-plan discounts, contract upgrade offers and loyalty benefits.
3. Investigate fiber customer churn by reviewing service reliability, installation experience, pricing, technical complaints, speed expectations and competitor offers.
4. Optimize the payment experience through payment education, easier payment options, auto-payment campaigns and targeted retention messaging for electronic check users.
5. Build a churn-risk segmentation strategy that combines attributes such as new customer, month-to-month contract and fiber optic service into a high-priority retention segment.

Conclusion

This analysis shows that churn is heavily concentrated among new customers, month-to-month subscribers and fiber optic users, with an overall churn rate of 26.5% against total customers of 7,043. Tenure, contract type, internet service and payment method each provide a meaningful signal for identifying at-risk customers. The interactive Power BI dashboard brings these signals together into a single tool, enabling stakeholders to prioritize retention efforts where they will have the greatest impact — particularly within the first 12 months of the customer lifecycle.